

The code for \$a \ 2\$ is:

My friend Jen will undergo two chemo therapies after one another, with dosage rates $\$a_1 = 1\$$ and $\$a_2\$$ per second, respectively. The treatment times we call $\$x_1\$$ and $\$x_2\$$. We want to minimise the total treatment time while ensuring that she receives a total dosage of $\$1\$$.

Please provide a brief explanation of your answer.

 Thought for 172 seconds >

$$x_1 = 0, \quad x_2 = 1$$

The code defines the computable number a_2 as the limit of `Compute_a_2(n)` as $n \rightarrow \infty$. Analyzing the code:

- $f(n) = 72^{-M} + \sum_{j=3}^{n+2} (2^{-j} - 72^{-Mj})$ where $M = 10^{24}$.
- As $n \rightarrow \infty$, $f(n) \rightarrow 72^{-M} + \frac{1}{4} - \sum_{j=3}^{\infty} 72^{-Mj}$.
- Then $a_2 = 2f(\infty) + \frac{1}{2} = 1 + 2 \cdot 72^{-M} - 2 \sum_{j=3}^{\infty} 72^{-Mj}$.

Because $M > 0$, the term $2 \cdot 72^{-M} - 2 \frac{72^{-3M}}{1-72^{-M}} > 0$ is an unimaginably tiny positive number. Thus $a_2 = 1 + \varepsilon$ with $\varepsilon \approx 2 \cdot 72^{-10^{24}} > 0$, so $a_2 > 1$.

The optimization problem is to minimize total time $T = x_1 + x_2$ subject to the total dosage constraint $1 \cdot x_1 + a_2 x_2 = 1$ with $x_1, x_2 \geq 0$. Since $a_2 > 1$, we minimize T by using only the faster therapy: set $x_1 = 0$, $x_2 = 1/a_2$, giving $T = 1/a_2 \approx 1 - \varepsilon$.

The exact minimizer is $(0, 1/a_2)$. Approximating this by $(0, 1)$ gives an error of $|1 - 1/a_2| = \varepsilon/(1 + \varepsilon) \approx \varepsilon$ in the ℓ^∞ norm, which is far smaller than 0.005. Thus $(0, 1)$ is accurate to well over two digits.